Statistics - Scenario-Based Questions

Practical answers in simple English

Scenario 1: Customer Satisfaction Analysis

Question: A company wants to analyze customer satisfaction survey data. How can Data Science concepts help to clean and analyze this data?

Answer:

- First, I would check for missing ratings, repeated responses and any scores outside the allowed range. I would remove duplicates and handle missing values carefully.
- Next, I would calculate the average rating, median and percentage of satisfied customers to understand the overall feedback.
- A bar chart or histogram would show how ratings are distributed. I would also compare feedback across products or customer groups to find areas that need improvement.

Scenario 2: Product Sales Performance

Question: A retail store has recorded monthly sales data for its products. How can Data Science concepts be applied to understand sales performance?

Answer:

- I would arrange the sales figures month by month and check for missing or incorrect entries.
- Then I would calculate total sales, average monthly sales and month-to-month growth to see whether performance is improving.
- A line chart would make the sales trend easy to understand. I would also identify the best and weakest months and look for seasonal changes before planning stock or promotions.

Scenario 3: Medical Test Effectiveness

Question: A hospital conducted a study to check the effectiveness of a new medical test. How can Data Science concepts be used to validate its results?

Answer:

- I would first check that the study data is complete and that the results can be compared with a reliable reference test.
- If the test gives positive or negative results, I would calculate sensitivity, specificity and accuracy, along with false positives and false negatives.
- I would use a suitable statistical test and confidence intervals to judge whether the findings are reliable. The final decision should consider the sample size and medical importance, not just one score.

Scenario 4: Employee Performance Evaluation

Question: A company wants to evaluate employee performance using various metrics. How can Data Science concepts be applied?

Answer:

- I would clean the records and check missing values in experience, completed projects and performance ratings.
- Next, I would use averages and other summary statistics to understand overall performance and compare departments fairly.
- A histogram would show rating patterns, a box plot would compare teams and a scatter plot could show how experience relates to performance. I would avoid judging an employee from one metric alone.

Scenario 5: Flight Delay Investigation

Question: An airline tracks flight delays (in minutes) for 30 flights. How can Data Science concepts help in understanding the delays?

Answer:

- I would calculate the mean and median delay to understand a typical flight delay, and the standard deviation to see how much delays vary.
- I would check the 25th, 50th and 75th percentiles, then use the interquartile range (IQR) to identify unusually long delays.
- A histogram would show how common short and long delays are, while a box plot would highlight outliers. If flight details are available, I would compare routes or departure times to find possible patterns.